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## MODEL SELECTION AND VALIDATION REPORT ( remove\_10\_biomass model )

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## OBSERVATIONS:1651

## SANITY: OK - all models in the selection process passed sanity checks.

## SELECTED RANDOM TERM: spatiotemporal\_random\_fields\_ar1

## MODEL SUMMARY ( remove\_10\_biomass model )

Spatiotemporal model fit by REML ['sdmTMB']

Formula: fixed\_formula

Time column: year

Family: lognormal(link = 'log')

Conditional model:

|                 | coef.est | coef.se |
|-----------------|----------|---------|
| (Intercept)     | 4.88     | 0.17    |
| remove_10_tric  | 0.61     | 0.03    |
| remove_10_teve  | -0.58    | 0.03    |
| sbt_mean        | 0.40     | 0.09    |
| sbt_sd          | 0.00     | 0.05    |
| sst_sd          | 0.03     | 0.04    |
| log_chla_mean   | 0.07     | 0.04    |
| depth           | -0.11    | 0.05    |
| log_sum_fishing | 0.06     | 0.03    |

Dispersion parameter: 1.03

Spatiotemporal AR1 correlation (rho): 0.90

Matérn range: 216.22

Spatiotemporal marginal AR1 SD: 0.81

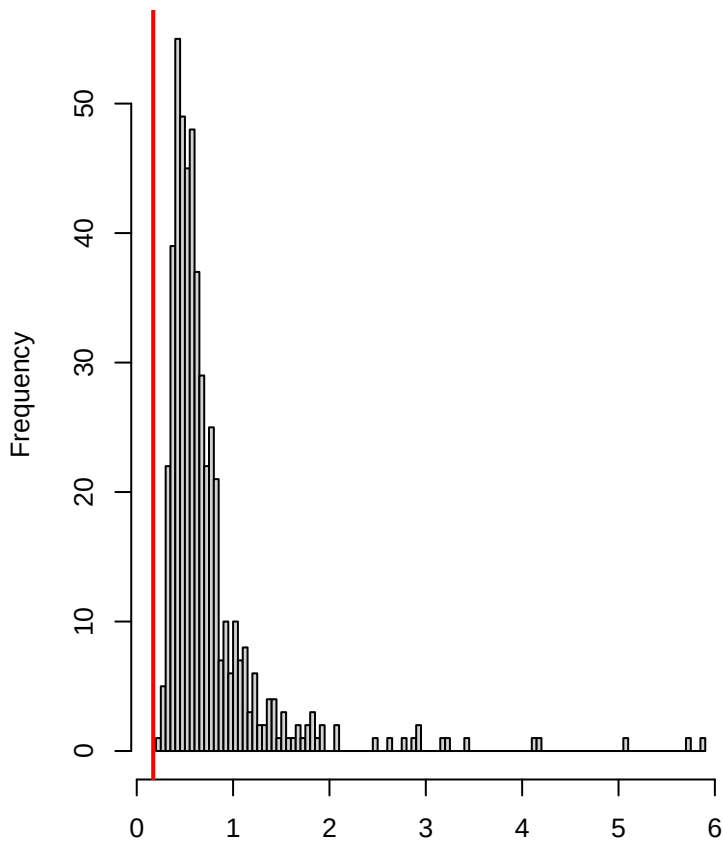
REML criterion at convergence: 9657.257

See ?tidy.sdmTMB to extract these values as a data frame.

## SELECTION STATISTICS

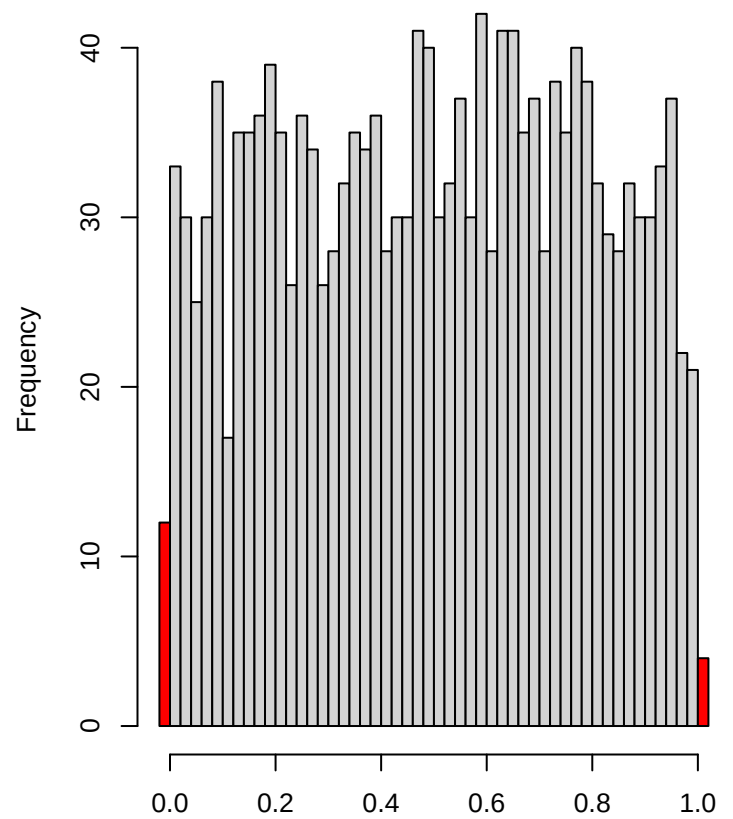
|                                                  | df | AIC      | delta_AIC  |
|--------------------------------------------------|----|----------|------------|
| spatiotemporal_random_fields_ar1                 | 13 | 19340.51 | 3.396547   |
| spatiotemporal_random_fields_ar1_with_anisotropy | 15 | 19343.91 | 69.680162  |
| spatial_random_fields                            | 12 | 19413.59 | 3.133584   |
| spatial_random_fields_with_anisotropy            | 14 | 19416.72 | 60.273083  |
| spatiotemporal_random_fields_iid                 | 12 | 19477.00 | 2.987891   |
| spatiotemporal_random_fields_iid_with_anisotropy | 14 | 19479.99 | 134.192032 |
| error_only                                       | 10 | 19614.18 | NA         |

DHARMa nonparametric dispersion test via sd of residuals fitted vs. simulated



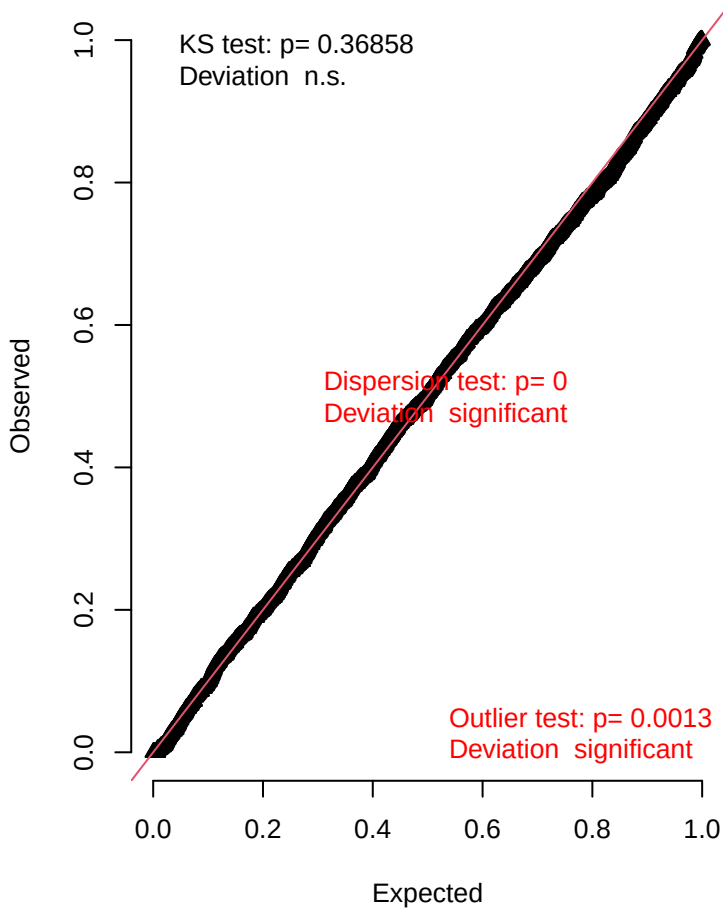
Simulated values, red line = fitted model. p-value (two.sided)

Outlier test significant

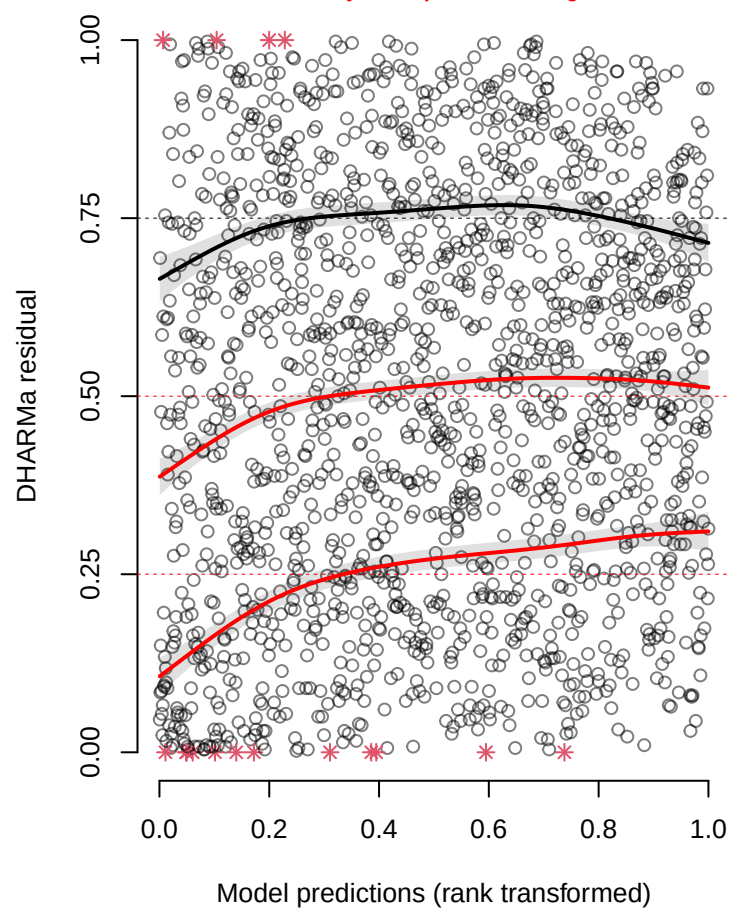


Residuals (outliers are marked red)

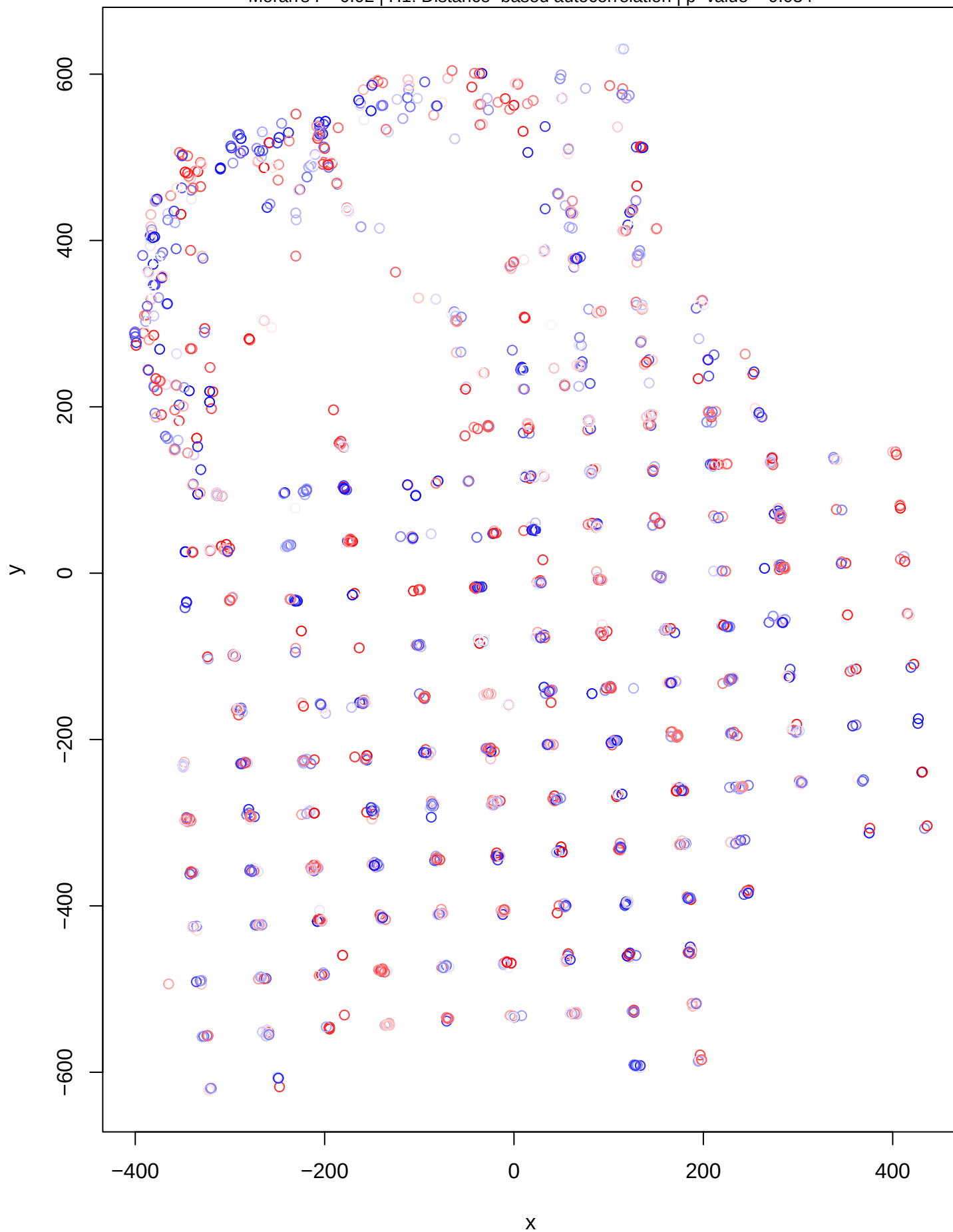
QQ plot residuals



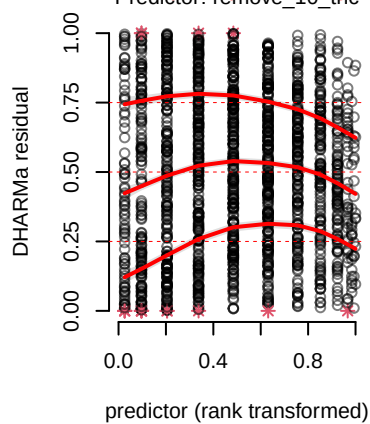
DHARMa residual vs. predicted  
Quantile deviations detected (red curves)  
Combined adjusted quantile test significant



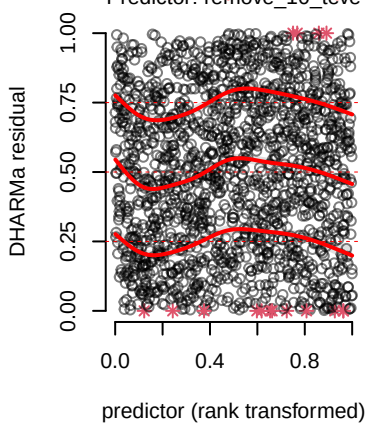
**DHARMA Moran's I test for distance-based autocorrelation**  
Moran's I = 0.02 | H1: Distance-based autocorrelation | p-value = 0.034



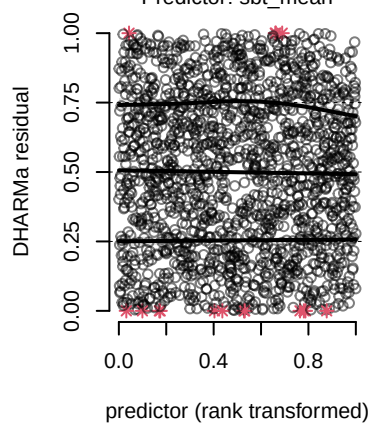
DHARMA residual Residual vs. predictor  
Quantile deviations detected (red curves);  
Combined adjusted quantile test significant



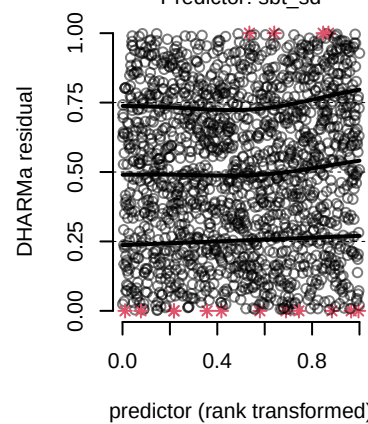
DHARMA residual Residual vs. predictor  
Quantile deviations detected (red curves);  
Combined adjusted quantile test significant



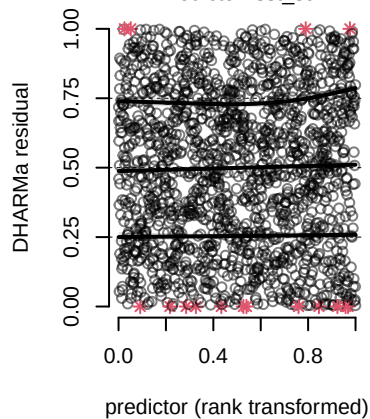
DHARMA residual Residual vs. predictor  
No significant problems detected  
Predictor: sbt\_mean



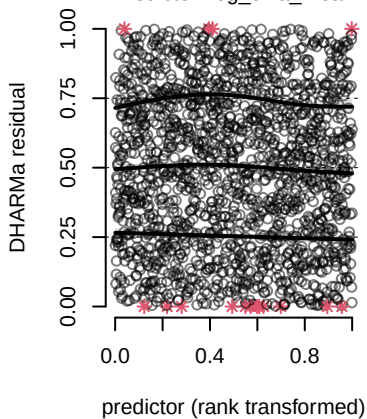
DHARMA residual Residual vs. predictor  
No significant problems detected  
Predictor: sbt\_sd



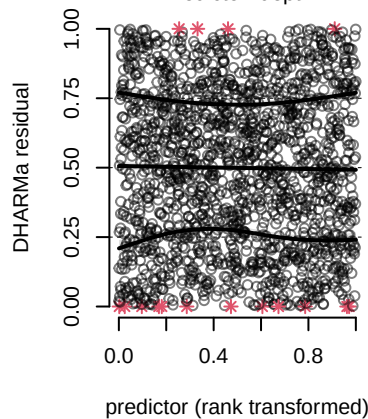
DHARMA residual Residual vs. predictor  
No significant problems detected  
Predictor: sst\_sd



DHARMA residual Residual vs. predictor  
No significant problems detected  
Predictor: log\_chla\_mean



DHARMA residual Residual vs. predictor  
No significant problems detected  
Predictor: depth



DHARMA residual Residual vs. predictor  
No significant problems detected  
Predictor: log\_sum\_fishing

